

CHAZ HYSENI

Dallas, TX, USA | +1-203-5082777 | chaz.hyseni@gmail.com | chazhyseni.github.io

Professional Summary

I am an Evolutionary Biologist and Population Geneticist with 20 years of experience in molecular techniques and statistical analyses, with extensive application of machine learning and deep learning methods throughout my career. Currently Senior Computational Scientist focused on AI/ML applications for genotype-phenotype associations in de-extinction research. Proven expertise in training large language models adapted for DNA sequences, implementing TensorFlow, Keras, and PyTorch pipelines for regulatory element prediction, and developing novel algorithms for cross-species genomic analysis. Strong background in translating evolutionary biology questions into machine learning problems and deploying computational solutions for complex biological systems.

Work History

Senior Computational Scientist Nov 2022 - Present

Colossal Biosciences – Dallas, TX

- Computational De-extinction Research: Determining genetic underpinnings of phenotypic evolution across vertebrates as part of de-extinction mission, applying evolutionary biology expertise to solve complex genotype-phenotype association problems
- AI/ML Pipeline Development: Designed and deployed large language models adapted for DNA sequence analysis using TensorFlow, Keras, and PyTorch, with specialized applications to regulatory genomics and cross-species trait prediction
- Algorithm Innovation: Developed novel computational approaches integrating evolutionary theory with deep learning methods for genomic feature prediction from pangenomes, including containerized deployment and GPU optimization
- Research Leadership: Coordinating computational aspects of de-extinction research, translating evolutionary biology questions into machine learning frameworks, and managing cross-functional collaborations

ORISE Postdoctoral Fellow Aug 2021 - Feb 2023

U.S. Forest Service – Oxford, MS

- Machine Learning Applications: Applied Extreme Learning Machines (ELM) and physics-informed neural networks for temperature modeling to assess climate change impacts on freshwater biodiversity, integrating environmental data with species distribution models
- Advanced Spatial Modeling: Integrated machine learning with traditional ecological methods for climate change impact modeling on freshwater biodiversity, implementing Bayesian spatial interpolation using Integrated Nested Laplace Approximation with Stochastic Partial Differential Equations for uncertainty quantification

Postdoctoral Researcher Aug 2020 - Aug 2021

Uppsala University – Uppsala, Sweden

- Performed genomic library preparation, integration and analysis of genetic and epigenetic data
- Applied machine learning and Bayesian statistics (R and Python) to study urban environmental effects on genetic variation and DNA methylation in invertebrate populations
- Implemented genetic algorithms for optimization of resistance surfaces in connectivity modeling, utilizing circuit theory-based approaches and omnidirectional connectivity analysis to map landscape corridors enabling species movement and population connectivity
- Investigated community genetics and connectivity patterns in urban pond metacommunities using spatial optimization algorithms and resistance-based modeling frameworks

Graduate Research Scientist Jan 2014 – May 2020

University of Mississippi – Oxford, MS

Designed molecular protocols and analytical workflows for genetic and epigenetic data integration

- Applied R and Python for spatial genetic analyses using machine learning and Bayesian statistics to analyze environmental influences on DNA methylation
- Implemented Random Forest, XGBoost, and neural networks for environmental modeling of biodiversity patterns and habitat suitability across landscape gradients
- Developed novel metrics for functional landscape connectivity across multiple organizational scales using spatial modeling approaches

Graduate Research Assistant Aug 2012 – Aug 2013

Cornell University – Ithaca, NY

- Conducted eastern oyster conservation research analyzing spatio-temporal genetic variation and gene-environment associations
- Performed genomic library preparation and optimized bioinformatic pipelines for genomic data analysis

Research Assistant Jun 2007 – Aug 2012

Yale University – New Haven, CT

- Performed molecular work and statistical analyses on two major projects: 1) Galápagos giant tortoise conservation genetics (phylogeography, speciation, hybridization), and 2) spatial and temporal patterns of genetic variation in disease vectors (tsetse) using Bayesian spatial modeling approaches
- Used R, Python, Linux, and Unix to write data analysis scripts and run bioinformatic pipelines, implementing Integrated Nested Laplace Approximation with Stochastic Partial Differential Equations for spatial analysis
- Trained staff and students in molecular techniques and data analysis at the DNA Analysis Facility

Education

Ph.D.: Biological Science 2014 – 2020

University of Mississippi – Oxford, MS

Dissertation: Evolutionary history of subterranean termites in the geographic and ecological context of the Appalachian Mountains in the United States

B.A.: Environmental Studies 2002 – 2007

Yale University – New Haven, CT

Technical Expertise

Core Biology Evolutionary biology, population genetics, molecular techniques, next-generation sequencing, bioinformatics, phylogeography

Machine Learning & AI TensorFlow, Keras, PyTorch, large language models for DNA sequences, Random Forest, XGBoost, neural networks, Extreme Learning Machines, physics-informed neural networks, deep learning pipelines, genetic algorithms for spatial optimization

Programming & Statistics Python, R, Linux/Unix, spatial statistics, Bayesian inference, Integrated Nested Laplace Approximation with Stochastic PDEs, multivariate analysis, circuit theory-based connectivity modeling

Research Leadership Cross-disciplinary coordination, algorithm development, computational pipeline design, team management